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Resting-state functional connectivity alterations in adult ADHD: A systematic review of rs-fMRI studies

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ABSTRACT

Background: Attention-Deficit/Hyperactivity Disorder (ADHD) often persists into adulthood and causes significant functional impairments. Resting-state functional MRI (rs-fMRI) offers valuable insights into intrinsic brain connectivity; however, connectivity (FC) alterations specific to adults with ADHD remain unclear. This systematic review aimed to synthesize rs-fMRI studies comparing adults with ADHD and healthy controls, identify consistent FC patterns, and assess methodological quality.

Methods: Following PRISMA guidelines, PubMed, Embase, and Scopus were searched for rs-fMRI studies involving adults diagnosed with ADHD per DSM-IV/DSM-5 criteria. Eligible studies examined FC differences between adults with ADHD and healthy controls. Data extraction included study characteristics, imaging parameters, and FC methodologies. Study quality was assessed using the Newcastle-Ottawa Scale.

Results: Eight studies (n = 529; 264 ADHD, 265 controls) met inclusion criteria. FC analyses employed Seed-Based Analysis (SBA), Independent Component Analysis (ICA), and graph theory methods. Despite methodological heterogeneity, consistent FC alterations were observed: increased FC in the default mode (DMN), visual (VN), and central executive (CEN) networks, and decreased FC in the ventral attention (VAN) and somatomotor (SMN) networks. Alterations were predominantly left-hemispheric.

Conclusion: Adults with ADHD exhibit distinct rs-FC disruptions, mainly involving DMN, VAN, CEN, and SMN. Standardized analytic approaches and subtype-specific analyses are needed for improved clinical relevance.

Abbreviations: ADHD, Attention-Deficit/Hyperactivity Disorder; FC, Functional Connectivity; rs-fMRI, resting state functional MRI; SBA, Seed Based Analysis; ICA, Independent Component Analysis; BOLD, blood-oxygen-level-dependent; RSNs, Resting State Networks; DPARSF, Data Processing Assistant for Resting-State fMRI; SPM, Statistical Parametric Mapping; DPABI, Data Processing & Analysis of Brain Imaging; FSL, the FMRIB Software Library; GIFT, Group ICA Of fMRI Toolbox; AFNI, Analysis of Functional Neuro-Images; 3T, 3 Tesla; SD, Standard Deviation; DMN, Default Motor Network; DAN, Dorsal Attention Network; CER, Cerebellum; mPFC, medial Pre-Frontal Cortex; PCC, Posterior Cingulate Cortex; LP, Lateral Parietal Cortex; AI, Anterior Insula; MI, Middle Insula; PI, Posterior Insula; RSC, Retro-Splenial Cortex; FG, Fusiform Gyrus; LG, Lingual Gyrus; IPS, Inferior Parietal Sulcus; PreCG, PreCental Gyrus; PoCG, PostCentral Gyrus; MCC, Mid-Cingulate Cortex; STG, Superior Temporal Gyrus; ITG, Inferior Temporal Gyrus; SOG, Superior Occipital Gyrus; MTG, Middle Temporal Gyrus; CEN, Central Executive Network; VN, Visual Network; SMN, Sensory Motor Network; VAN, Ventral Attention Network; ROI, Region of Interest; DPP, Double Ping-Pong; PPO, Ping-Pong Output; PPR, Ping-Pong Receive; HIP, Hippocampus; HG, Helsch's Gyrus; IFG, Inferior Frontal Gyrus; CAL, Calcarine; PHG, Para-Hippocampal Gyrus; OFC, Orbito-Frontal Cortex; SFG, Superior Frontal Gyrus; BG, Basal Ganglia; SCN, Subcortical Network; GP, Global Palidus, FPN, Fronto Parietal Network; AG, Angular Gyrus; laPFC, Left Anterior Pre-Frontal Cortex; MFG, Middle Frontal Gyrus; RG, Rectal Gyrus; MOG, Middle Occipital Gyrus; ACC, Anterior Cingulate Cortex; SPL, Superior Parietal Lobule; IOG, Inferior Occipital Gyrus; dmPFC, dorsomedial Pre-Frontal Cortex.

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1. Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a common neurodevelopmental condition characterized by persistent symptoms of inattention, hyperactivity, and impulsivity. While historically considered a childhood disorder, ADHD frequently persists into adulthood, affecting approximately 2.5-5% of adults worldwide (Kessler et al., 2006; Kosheleff et al., 2023). Adult ADHD is associated with considerable functional impairments across educational, occupational, and interpersonal domains. Despite increasing clinical awareness, the neurobiological underpinnings of adult ADHD remain incompletely understood (Alexander and Farrelly, 2018).

Resting-state functional magnetic resonance imaging (rs-fMRI) has emerged as a powerful, non-invasive neuroimaging technique to examine the brain's intrinsic functional architecture. By measuring spontaneous fluctuations in the blood-oxygen-level-dependent (BOLD) signal, rs-fMRI enables the identification of functional connectivity (FC) patterns between distributed brain regions in the absence of task-related activity (Tang et al., 2022). This modality has been instrumental in revealing atypical connectivity in several large-scale resting state networks (RSNs) implicated in ADHD, including the default mode network (DMN), central executive network (CEN), ventral attention network (VAN), dorsal attention network (DAN), visual network (VN) and sensory motor network (SMN). This approach is particularly advantageous in ADHD, as task-based evaluation can be difficult in children and may introduce confounding factors related to task performance. fMRI has become a key tool in elucidating the neural underpinnings of ADHD with various analytical approaches offering distinct perspectives on brain connectivity. Independent Component Analysis (ICA), a data-driven method, is commonly used to identify intrinsic connectivity networks without requiring predefined regions. Seed-Based Analysis (SBA), in contrast, is hypothesis-driven and measures FC between a selected seed region and the rest of the brain, allowing for targeted investigation of circuits implicated in ADHD. Region of Interest (ROI)-to-ROI analysis extends this by examining FC between multiple predefined brain regions to map broader network dysfunctions. (Rosazza et al., 2012) Furthermore, graph-theoretical approaches conceptualize the brain as a network, enabling the quantification of topological properties such as integration, segregation, and centrality—metrics increasingly applied to characterize altered network organization in ADHD. Together, these analytical strategies provide a comprehensive framework for understanding the distributed and network-based nature of ADHD-related brain dysfunction.

While there is a growing body of literature examining fMRI connectivity in ADHD, the majority of studies have focused on children and adolescents or task-based fMRI (Cubillo and Rubia, 2010; Cortese et al., 2021; Tamon et al., 2024; Yu et al., 2022; Kim et al., 2023; Cortese et al., 2012; Saletin et al., 2019; Noordermeer et al., 2016; Lei et al., 2015; Gao et al., 2021). Findings of rs-fMRI in adult ADHD are more limited and often inconsistent, reflecting heterogeneity in study populations, imaging methodologies, and analytic approaches. Furthermore, differences in brain networks between children and adults suggest that FC alterations in adult ADHD may represent persistent or compensatory neural mechanisms distinct from those observed in younger individuals (Hong et al., 2024).

Comprehensive synthesis systematically evaluating the FC findings from rs-fMRI studies specifically focused on adults with ADHD is lacking. This systematic review addresses this gap by critically appraising and summarizing existing literature comparing rs-fMRI connectivity in adults with ADHD to healthy control (HC) groups. Our objective is to summaries existing literature, identify consistent patterns of altered brain connectivity associated with adult ADHD.

2. Methods

2.1. Inclusion criteria

Studies were included if they met all the following criteria: a) original, peer-reviewed articles published in English; b) adult patients (aged ≥18 years) diagnosed with ADHD based on DSM-IV or DSM-5 criteria with a comparable group of HCs; c) use of rs-fMRI; and d) application of any FC analysis method using rs-fMRI. There were no limitations regarding the nationality, race, sex of the participants.

2.2. Exclusion criteria

Studies were excluded if they met any of the following criteria: a) animal studies; b) children and adolescents (<18 years); c) absence of HCs, d) insufficient information on FC changes; e) meeting abstracts, comments, summaries, protocols, letters and guidelines; and f) articles not written in English. In case of overlapping cohorts; the most recent study was included.

2.3. Search strategy

The systematic review was conducted accordance with a predefined protocol (PROSPERO 2025 CRD420251041135). It was designed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines. A systematic search was performed of PubMed/MEDLINE, Embase, and Scopus to identify studies investigating rs-fMRI in neurodegenerative or psychiatric disorders from inception until February 25, 2025, by two independent reviewers (J.K. and B.M.). The detailed search string for each database is provided in the supplementary material.

Search results were imported into Covidence, where duplicates were automatically screened and deleted. Two authors (J.K. and B.M.) independently screened the titles and abstracts of the remaining studies. Screening results were examined by a third reviewer (L.P.L., neuroradiologist, 15 years of experience), and all discordant evaluations were resolved via discussion to reach a consensus. As this review is a part of larger systematic review, studies involving adult ADHD populations were further categorized. A complete flow chart of the process is shown in the supplementary material.

2.4. Data Extraction

Data extraction was performed independently by two investigators (J.K. and B.M.). For each article, we extracted: a) patient and control demographics b) MRI scanner strength, c) software used, d) FC analysis method, e) Number and location of seeds (if seed-based), f) Direction of FC changes (increase/decrease), and g) regions with FC difference, and h) Brain network involved

For the analysis of the most frequently involved brain networks, only whole-brain studies were included. To minimize selection bias, studies restricted to predefined regions or specific networks were excluded when reporting results from whole-brain investigations.

2.5. Quality Evaluation

Studies were appraised with the Newcastle-Ottawa Scale (NOS), scored on a 0 to 9 scale based on patient selection, comparability of groups and outcome assessment. Studies with a total NOS score of 7 or higher (out of 9) were considered high quality. Discrepancies were resolved through discussion by a third independent reviewer (L.P.L.).

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3. Results

3.1. Study selection and overview

The study selection process and reasons for exclusion during full text screening are described in supplementary material. Out of 4,405 potentially eligible citations, 8 studies met inclusion criteria. Study characteristics are reported in Table 1.

In total, the included studies involved 529 participants: 264 adults with ADHD (SBA = 71; ICA = 65; GBA = 128) and 265 HCs (SBA = 73; ICA = 65; GBA = 127).

3.2. Seed-based/ROI to ROI studies

Hwang et al. (Hwang et al., 2025) reported increased FC predominantly in bilateral lateral parietal (LP) cortex regions when seeded from the posterior cingulate cortex (PCC). Kucyi et al. (Kucyi et al., 2015) investigated cerebellum-cerebral connectivity using a cerebellar seed and observed increased FC in multiple regions, including areas within the DMN, SMN, VAN, VN, and CEN. Zhao et al. (Zhao et al., 2017) examined whole brain FC using six insular subregions as seeds and found increased FC particularly in the right precuneus alongside decreased FC in several regions, spanning multiple networks. (Supplementary material)

3.3. Non-seed based studies

Zhang et al. (Zhang et al., 2021) employed ICA to assess FC between eight RSNs, noting increased connectivity in the right fronto-parietal network (FPN) and auditory regions, while decreased connectivity was observed within the anterior medial prefrontal cortex (mPFC) and anterior cingulate cortex (ACC) compared to HCs. Wu et al. (Wu et al., 2023) applied motif analysis as a graph-theoretical metric and identified significant connectivity alterations involving all major large-scale networks. Pretus et al. (Pretus et al., 2019) evaluated stepwise FC changes using integration as a graph metric, observed increased FC in the left calcarine (CAL) and right lingual gyrus (LG), while orbitofrontal cortex (OFC) showed reduced FC in ADHD. Hong et al. (Hong et al., 2017) reported increased nodal centrality in the VN and decreased centrality across multiple areas within the CEN, VAN, SMN, and sub-cortical network (SCN). Fan et al. (Fan et al., 2019) noted increased clustering co-efficient and path length within DMN. (Supplementary material)

3.4. Illustrative brain map of replicated neuroimaging findings

Fig. 1 summaries regions with altered FC. we identified regions with consistently increased FC including left AG in the DMN, left LG and medial occipital gyrus in the VN, left superior frontal gyrus (SFG) within the CEN. The left anterior pre-frontal cortex (laPFC) and insula within the VAN also showed increased FC. Decreased FC was observed in the left pre-central gyrus (PreCG) and right superior temporal gyrus (STG) within the SMN, right putamen in the SCN, bilateral Insula in the VAN (Fig. 1; Supplementary material).

Overall, increased FC was predominantly observed across DMN, VN, and CEN; while decreased FC was primarily seed in the VAN and SMN. Considering only whole brain studies, most consistently altered networks includes the DMN, VN, and SMN (71%) (Fig. 2).

3.5. Quality of studies

Study quality was appraised using the Newcastle-Ottawa Scale (NOS), as illustrated in Fig. 3. Eight case-control studies were included in the quality evaluation. Of these, three were rated as having a low risk of bias, while five demonstrated a moderate risk. No studies were classified as high risk. The observed sources of bias primarily stemmed from incomplete reporting of data and suboptimal control group definition or selection procedures. Overall, the methodological quality of the included studies was deemed acceptable for synthesis.

4. Discussion

Our study sought to evaluate FC alterations in adults with ADHD using rs-fMRI, an area that remains underexplored compared to pediatric populations. Prior studies described pattern of widespread but heterogeneous FC alterations across multiple large-scale RSNs. We identified two prior meta-analyses with separate analyses on adult ADHD (Cortese et al., 2012; Lei et al., 2015), both task-based. Additionally, two rs-fMRI-based meta-analyses were identified, one focusing on local FC measures in adolescent (Shu et al., 2025), another including

Table 1
Characteristics of included studies.

References	Patient (M/F)	Mean Age Patient (SD)	Healthy Control (M/F)	Mean Age Control (SD)	Strength of MRI	Software	Method	No. of seeds	Seeds
Hwang, 2025 (Hwang et al., 2025)	20(14/6)	23.7(2.7)	20(15/5)	24.5(2.7)	3T	DPARSF, SPM12	ROI to ROI	8 (4 - DMN, 4 - DAN)	DMN: mPFC, B_LP, PCC; DAT: B_FEF and B_IPS
Zhao, 2017 (Zhao et al., 2017)	28(15/13)	27.07(5.48)	30(17/13)	25.92(3.77)	3T	DPABI	Seed based	6 (Insula)	B_AI, B_MI, B_PI
Kucyi, 2015 (Kucyi et al., 2015)	23(10/13)	24.3(3.9)	23(8/15)	24.2(2.9)	3T	FSL	Seed based	1 (CER) 2 (CER)	B_Crus I
Zhang, 2021 (Zhang et al., 2021)	41(20/21) 24(19/5)	32.85(10.89) 34.75(9.712)	42(21/21) 23(14/9)	32.45(9.05) 34.89(8.387)	3T	GIFT	ICA		
Wu, 2023 (Wu et al., 2023)	42(21/21)	32.71(10.47)	50(27/23)	32(8.96)	3T	DPABI	Graph based (Motif based)		
Pretus, 2019 (Pretus et al., 2019)	31(15/16)	35.4(9.9)	22(10/12)	30.4(5.8)	3T	SPM12, AFNI	Graph based (integration)		
Hong, 2017 (Hong et al., 2017)	31(19/12)	26.77(2.54)	31(21/10)	24.94(1.41)	3T	AFNI, FSL	Graph based (centrality)		
Fan, 2019 (Fan et al., 2019)	24	NA	24	NA	3T	AFNI	Graph based (Segregation)		

DPARSF - Data Processing Assistant for Resting-State fMRI; SPM - Statistical Parametric Mapping; DPABI - Data Processing & Analysis of Brain Imaging; FSL - the FMRIB Software Library; GIFT - Group ICA Of fMRI Toolbox; AFNI - Analysis of Functional Neuro-Images; 3T - 3 Tesla; SD - Standard Deviation; DMN - Default Motor Network; DAN - Dorsal Attention Network; CER - Cerebellum; mPFC - medial Pre-Frontal Cortex; PCC - Posterior Cingulate Cortex; LP - Lateral Parietal Cortex; AI -Anterior Insula; MI - Middle Insula; PI - Posterior Insula; FEF - Frontal Eye Field; IPS - Inferior Parietal Sulcus; SD - Standard Deviation.

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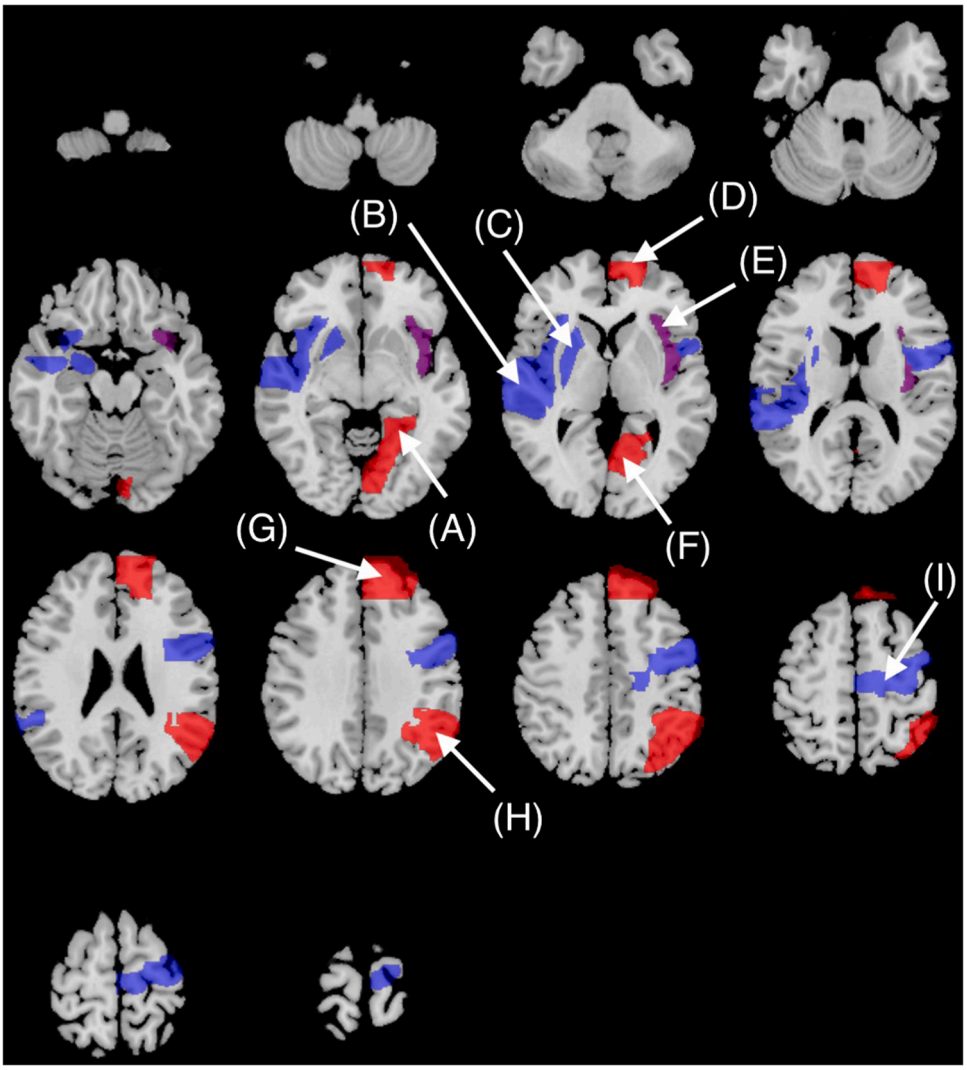


Fig. 1. Overlapping region with altered FC; Blue area indicates decreased FC; Red area indicated increased FC: (A) Left PCC/Precuneus; (B) Right Superior Temporal Gyrus; (C) Right Putamen (D) Left Anterior PFC (E) Left Insula (F) Left Lingual Gyrus (G) Left MFG (H) Left Angular Gyrus (I) Left PreCentral Gyrus.

only seed based studies only without age-specific analyses. (Sutcubasi et al., 2020) We found no systemic-review or meta-analysis encompassing all types of FC changes in adult ADHD using rs-fMRI.

Despite heterogenous literature, we observed a consistent pattern of FC changes in adults with ADHD at rest. Regions with consistently altered FC included increased FC in the left AG, LG, laPFC and SFG (involving DMN, VN, and CEN). Decreased FC was observed in the right insula, right putamen, right STG, and left PreCG (involving VAN, SMN). The left insula (VAN) exhibited both increased and decreased FC across studies (Fig. 1; Supplementary material). Moreover, the calcarine sulcus, showed increased FC in non-seed-based studies (Zhang et al., 2021; Wu et al., 2023; Pretus et al., 2019; Fan et al., 2019); but not in seed-based analyses. laPFC FC alteration can be explained by prior literature showed dysfunction and delayed maturation of PFC is a key abnormality in ADHD (Soman et al., 2023). The lingual gyrus and visual cortical areas are essential for sensory integration, visual processing, and attentional control. VN Hyperconnectivity demonstrated in our analysis reflects involvement of occipital lobe in visual attention task in ADHD. (Li et al., 2025; Zarka et al., 2021) Insular cortex dysfunction in ADHD contributes to sensory integration deficits, emotional dysregulation, and impaired cognitive control, reflecting its role as a multimodal integrative hub within the VAN. (Fateh et al., 2022; Viering et al., 2021)

Our findings align with the prior rs-fMRI meta-analyses reporting FC changes in the left precuneus/PCC, left LG, left MFG, and laPFC. (Shu et al., 2025; Sutcubasi et al., 2020) This overlap reinforces the idea that rs-fMRI may capture distinct and consistent patterns compared to task-based approaches. Interestingly, these findings suggest dominant involvement of the left hemisphere, in contrast to previous task-based fMRI meta-analyses reporting right-hemisphere abnormalities in similar regions.

The most commonly involved regions with altered FC in prior task-based meta-analyses include the right sided PreCG, insula, and precuneus/PCC. (Tamon et al., 2024; Yu et al., 2022; Kim et al., 2023; Cortese et al., 2012; Saletin et al., 2019; Lei et al., 2015; Gao et al., 2021; Kowalczyk et al., 2022) In contrast, our review found altered FC in left hemisphere counterparts (PreCG, insula, and precuneus/PCC). This hemispheric discrepancy may reflect the nature of the imaging modality: task-based fMRI studies are inherently influenced by lateralization and handedness, often biasing activation toward the right hemisphere during attention and inhibition tasks. (Toga and Thompson, 2003; Szaflarski et al., 2006; Holland et al., 2007; Leckey and Federmeier, 2017; Tzourio-Mazoyer et al., 2020) In contrast, rs-fMRI captures intrinsic brain activity without task confounds, potentially offering a more stable and generalizable assessment of FC. Literature suggests hemispheric dominance evolves with age, with the left hemisphere becoming increasingly specialized for executive and cognitive control functions, while right hemisphere functions become more lateralized for visuospatial and attentional tasks. (Toga and Thompson, 2003;

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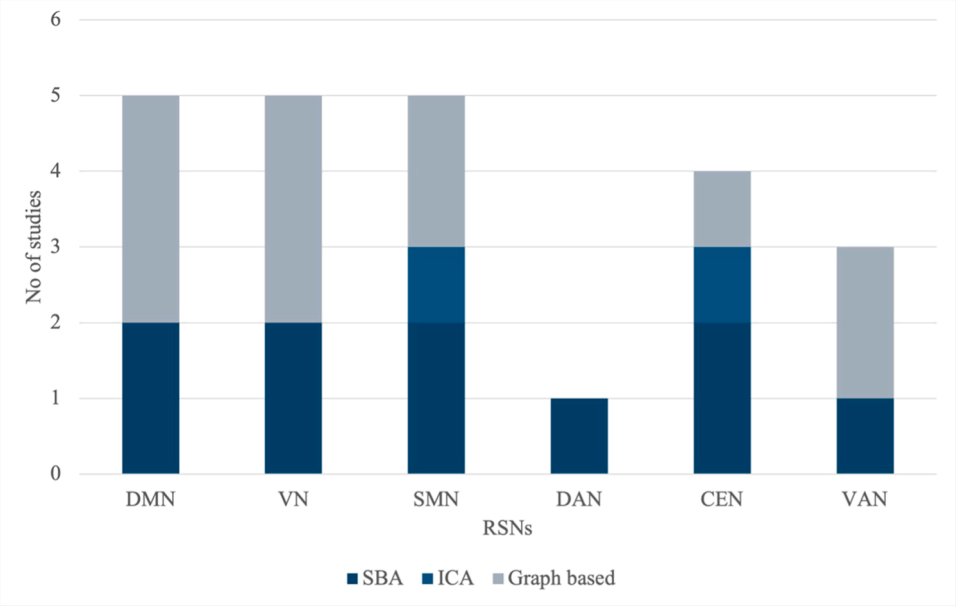


Fig. 2. Number of studies showed altered FC of RSNs; RSNs - Resting State Networks.

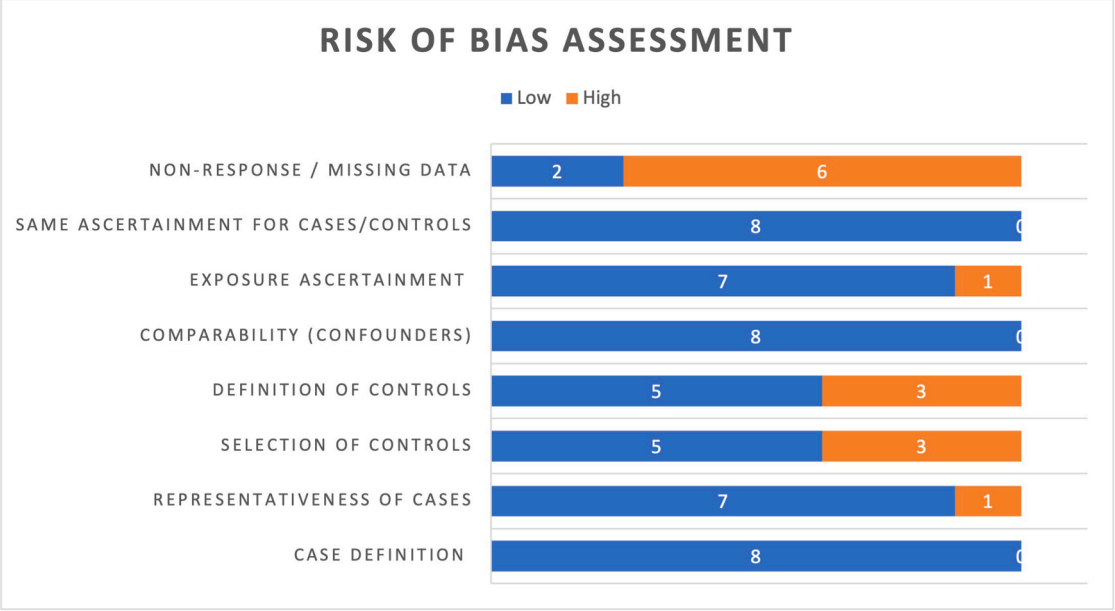


Fig. 3. Source of bias in case control studies (n=8).

Szaflarski et al., 2006; Holland et al., 2007; Leckey and Federmeier, 2017; Tzourio-Mazoyer et al., 2020) These developmental shifts may explain the hemispheric FC differences observed in our review.

Other regions with FC changes that align with our findings include the AG (Cortese et al., 2012); putamen (Tamon et al., 2024; Cortese et al., 2012; Lei et al., 2015); STG (Tamon et al., 2024; Yu et al., 2022; Cortese et al., 2012; Gao et al., 2021); MFG (Tamon et al., 2024; Kim et al., 2023; Cortese et al., 2012; Lei et al., 2015; Gao et al., 2021; Shu et al., 2025); and LG (Tamon et al., 2024; Shu et al., 2025). Prior studies have also reported FC changes in various other regions not consistently replicated across studies, such as the MOF, IPS, IFG, FG, thalamus, GP, amygdala, and caudate, highlighting high variability and the need for more uniform analytic approaches. Conversely, a few meta-analyses reported no significant FC changes. (Cortese et al., 2021; Toga and Thompson, 2003) Detailed findings from prior meta-analyses are summarize in supplementary material.

The heterogeneity observed across included studies is due to fundamental differences in FC analysis methodology. SBAs are hypothesis driven measure in which results are biases toward connectivity patterns involving pre-determined areas identified as a seed region, overlooking alterations in regions not selected a priori. (Joel et al., 2011) Similarly, ICA introduces variability through dimensionality estimation includes alteration in identification of brain network and their connectivity due to extracted components varied. (Joel et al., 2011) Graph-theoretical approaches are particularly sensitive to thresholding and the parcellation scheme utilized; different atlases and threshold values can result in divergent network topology metrics even from identical primary data. (Welton et al., 2015) Additionally, Motion correction strategies also differ across studies as individuals with ADHD are known to demonstrate higher rates of micro-movements during

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scanning, and inconsistent scrubbing or censoring thresholds may introduce noise that potentially affects results. (Epstein et al., 2007) These methodological sources of variability likely contribute to the inconsistent and sometimes divergent FC findings reported across studies in this review, and underscore the need for standardized acquisition and analysis pipelines in future rs-fMRI research.

4.1. Limitations

First, our results are based on a limited number of studies (n=8) with small sample sizes, which limits generalizability. Second, due to this limitation, our analysis relied on the identification of overlapping regions rather than statistical meta-analysis, which reduces the robustness of effect estimates. Third, the included studies employed heterogeneous FC analysis methods (e.g., SBA, ICA, graph theory), introducing variability in results and interpretation. Forth, medication status represents a significant potential confounder as stimulant medications are known to modulate BOLD signal and alter FC; included studies varied considerably in whether participants were medicated, unmedicated, or medication-naive, and stratified analyses was not performed by medication status. Fifth, comorbid psychiatric conditions (depression, anxiety disorders, and substance use disorders) are highly prevalent in adults with ADHD and can independently affect FC pattern; however, comorbidity profiles were rarely used as covariates in studies. Sixth, head motion during rs-fMRI acquisition is a well-established confound, and adults with ADHD tend to demonstrate higher micro-movements than neurotypical controls in scanner; differences in motion correction and censoring thresholds introduce bias in FC estimates. Finally, the age ranges and sex distributions affect resting-state network organization and is varied across included studies. Future studies should focus towards controlling these variables to improve comparability and interpretability of rs-fMRI findings in adult ADHD.

5. Conclusion

This systematic review synthesizes current evidence on rs-FC alterations in adults with ADHD, revealing widespread but heterogeneous disruptions across key brain networks predominantly involving left sided DMN, LN, and SMN. These hemispheric differences may reflect developmental shifts and/or task-based nature of earlier studies. Future studies should aim for larger, more diverse samples, incorporate longitudinal and multimodal designs, and stratify results based on clinical subtypes and comorbidities to explore the utility of resting-state imaging as a diagnostic and monitoring tool in adult ADHD.

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CRediT authorship contribution statement

Jay Kakadiya: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Beatriz A. de Macedo:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Neel Vora:** Writing – original draft, Methodology, Conceptualization. **Vidhi Dhaduk:** Writing – original draft, Methodology, Conceptualization. **Dhairya A. Lakhani:** Writing – review & editing, Supervision, Methodology. **Haris I. Sair:** Writing – review & editing, Supervision, Conceptualization. **Licia P. Luna:** Writing – review & editing, Supervision, Software, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.pscychresns.2026.112285.

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